

We believe in building our talent pool by inducting young professionals and grooming them into competent power professionals through various HR interventions.

We have developed a well-structured system for recruiting fresh engineering graduates and diploma holders at the executive level. We recruit people not only on the basis of qualification and skill repertoire but, more importantly, also for their attitude and cultural fit with the company. We even arrange training programmes for students of different technical institutes.”

—Neelam Kumar, executive director and head of the department-power electronics, Aplab



be pursued in the fields of technical marketing, design engineering, project management, and integration and servicing.

If you consider the application of power electronics, I must say it is present wherever there is a need for control and conversion of electric power. It may be the tiny converter of your mobile phone in the range of some milliwatts, or a giant one of hundreds of megawatts in a high-voltage direct-current (HVDC) transmission system. The main metric of power electronics is energy efficiency. And as a professional you are expected to acquire knowledge and skills needed to design practical power electronics systems. If you can achieve it, then control, instrumentation and maintenance of power systems will automatically be on your fingertips.

Largely, this is a technical field and so domain expertise and process knowledge are very important. Because of the technology-driven nature of this field, professionals must at all times keep themselves abreast of the latest improvements and technological up-gradations that are taking place instead of focusing only on their departmental

concerns. In addition to technical skills, recruiters are also looking for competencies such as interpersonal skills and drive for end results down the ranks. Don't get nervous. You are not expected to know everything but your learning ability should be high as a lot of training takes place on the job.

Powerful fundamentals

Keep in mind that power electronics is primarily concerned with the processing of electrical power using electronic devices. So the key element is the switching converter. All other devices are improved versions of it.

In general, a switching converter comprises three ports—power input port, power output port and control. Converters are even classified according to the nature of these ports. For example, in a simple DC-DC converter, the DC input voltage is converted into a DC output voltage by changing its magnitude. The device may perform some additional functions like rectification and inversion of voltage, thereby acquiring the names 'rectifier' and 'inverter', respectively. The third port, i.e., control, is required to produce a

well-regulated output voltage in presence of variations in the input and load currents.

I have already mentioned the metric of power process. High efficiency is a must for any processing system. The primary reason for this is not to save money on one's electricity bill or conserve energy. Rather, according to power rule, it is impractical to design a low-efficiency converter which can produce high voltage output.

The next thing that you need to know is the application of power electronics. A basic design is modified according to the need of the application. Let's consider an aerospace application of power electronics. In the power system of an earth-orbiting spacecraft, a solar array produces the main power bus voltage. A DC-DC converter converts the bus voltage into the regulated voltage required by the spacecraft payloads. Battery charge controllers interface the main power bus to the batteries; this controller may also have a DC-DC converter. The power systems of almost all spacecrafts and aircrafts follow this simple scheme.

On the other hand, in an electrically driven vehicle, batteries are charged by converters that draw high-factor sinusoidal current from a single-phase or a three-phase AC line. The batteries supply power to the variable-speed AC motors to propel the vehicle. The speed of the AC motor is controlled by the variation of the electrical input frequency. The inverter produces three-phase variable output to control the speed of the motor as well as the vehicle. In some applications, a DC-DC converter steps down the original voltage to a lower level according to the electronics need of the systems.

The list of examples may be endless. So instead of mugging up the nitty-gritty of individual systems, try to understand the function of the individual elements, so that you can use them to design the power system of your choice. You may put special emphasis on various common systems to recheck your understanding. A